



NCI Protocol #: 10323

Patient ID:

Date:

1 of 11

Patient Name:

Date Collected:

Biopsy Site: **Bone Marrow Aspirate**

Sample ID:

Telephone:

Tumor Content (%): **60%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Recurrent Plasma Cell Neoplasm**

MoCha ID:

Fax:

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IIC	<i>KRAS Q61E, KRAS L19F</i> Prognostic significance: None Diagnostic significance: None	None	None	3
III	<i>ATM D2016Y, ATM E2272K</i> Prognostic significance: None Diagnostic significance: None	None	None	3
IIC	<i>PALB2 R753*</i> Prognostic significance: None Diagnostic significance: None	None	None	3
III	<i>MSH6 C687F</i> Prognostic significance: None Diagnostic significance: None	None	None	2
III	<i>MRE11A L485F</i> Prognostic significance: None Diagnostic significance: None	None	None	2

Public data sources included in relevant therapies: FDA1, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Prevalent cancer biomarkers without relevant evidence based on included data sources

SETD2 Q1931, SETD2 S1505fs*

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.03(006). The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
MSH6	p.(C687F)	c.2060G>T	.	chr2:48027182	18.93%	NM_000179.2	missense
SETD2	p.(Q1931*)	c.5791C>T	.	chr3:47125479	20.16%	NM_014159.6	nonsense
SETD2	p.(S1505fs)	c.4514_4516delCTAinsT	.	chr3:47158183	11.13%	NM_014159.6	frameshift Block Substitution
MRE11A	p.(L485F)	c.1453C>T	.	chr11:94192621	10.94%	NM_005591.3	missense
ATM	p.(D2016Y)	c.6046G>T	.	chr11:108186589	18.07%	NM_000051.3	missense
ATM	p.(E2272K)	c.6814G>A	.	chr11:108196791	17.74%	NM_000051.3	missense
KRAS	p.(Q61E)	c.181C>G	COSM550	chr12:25380277	32.90%	NM_033360.3	missense
KRAS	p.(L19F)	c.57G>C	.	chr12:25398262	28.80%	NM_033360.3	missense
PALB2	p.(R753*)	c.2257C>T	.	chr16:23641218	50.85%	NM_024675.3	nonsense

Comments:
N/A

Signed By: _____

Biomarker Descriptions

ATM (ATM serine/threonine kinase)

Background: The ATM gene encodes a serine/threonine kinase that belongs to the phosphatidylinositol-3-kinase related kinases (PIKKs) family of genes that also includes ATR and PRKDC (also known as DNA-PKc)¹. ATM and ATR act as master regulators of DNA damage response. Specifically, ATM is involved in double-stranded break (DSB) repair while ATR is involved in single-stranded DNA (ssDNA) repair². ATM is recruited to the DNA damage site by the MRE11/RAD50/NBN (MRN) complex that senses DSB^{2,3}. Upon activation, ATM phosphorylates several downstream proteins such as the NBN, MDC1, BRCA1, CHK2 and TP53BP1 proteins⁴. ATM is a tumor suppressor gene and loss of function mutations in ATM are implicated in the BRCAness phenotype, which is characterized by a defect in homologous recombination repair (HRR), mimicking BRCA1 or BRCA2 loss^{5,6}. Germline mutations in ATM often result in

Biomarker Descriptions (continued)

Ataxia-telangiectasia, a hereditary disease also referred to as DNA damage response syndrome that is characterized by chromosomal instability⁷.

Alterations and prevalence: Recurrent somatic mutations in ATM are observed in 17% of endometrial carcinoma, 15% of undifferentiated stomach adenocarcinoma, 13% of bladder urothelial carcinoma, 12% of colorectal adenocarcinoma, 9% of melanoma as well as esophagogastric adenocarcinoma and 8% of non-small cell lung cancer^{8,9}.

Potential relevance: The PARP inhibitor, olaparib¹⁰ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes ATM. Consistent with other genes associated with the BRCAness phenotype, ATM mutations may aid in selecting patients likely to respond to PARP inhibitors^{5,11,12}. Specifically, in a phase II trial of metastatic, castration-resistant prostate cancer, four of six patients with germline or somatic ATM mutations demonstrated clinical responses to olaparib¹³. In 2022, the FDA granted fast track designation to the small molecule inhibitor, pidnarulex¹⁴, for BRCA1/2, PALB2, or other homologous recombination deficiency (HRD) mutations in breast and ovarian cancers.

KRAS (KRAS proto-oncogene, GTPase)

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival^{15,16,17}.

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. KRAS mutations are observed in up to 10-20% of uterine cancer, 30-35% of lung adenocarcinoma and colorectal cancer, and about 60% of pancreatic cancer⁹. The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61^{9,18,19}. Mutations at A59, K117, and A146 have also been observed but are less frequent^{8,20}.

Potential relevance: The KRAS inhibitor, sotorasib²¹, is approved (2021) for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). The FDA has granted breakthrough therapy designation (2021) to the small molecule inhibitor, adagrasib, for KRAS G12C positive in non-small cell lung cancer following prior systemic therapy²². The small molecular inhibitor, RO-5126766, was also granted breakthrough designation (2021) alone for KRAS G12V mutant non-small cell lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer²³. Additionally, onvansertib²⁴ was granted fast track designation (2020) for second-line treatment of patients with KRAS-mutated metastatic colorectal cancer (mCRC). The EGFR antagonists, cetuximab²⁵ and panitumumab²⁶, are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)²⁰. Additionally, KRAS mutations are associated with poor prognosis in NSCLC²⁷.

MSH6 (mutS homolog 6)

Background: The MSH6 gene encodes the mutS homolog 6 protein²⁸. MSH6 is a tumor suppressor gene that heterodimerizes with MSH2 to form the MutSa complex²⁹. The MutSa complex functions in the DNA damage recognition of base-base mismatches or insertion/deletion (indels) of 1-2 nucleotides²⁹. DNA damage recognition initiates the mismatch repair (MMR) process that repairs mismatch errors which typically occur during DNA replication. Mutations in MSH2 result in the degradation of MSH6³⁰. MSH6, along with MLH1, MSH2, and PMS2 form the core components of the MMR pathway²⁹. The MMR pathway is critical to the repair of mismatch errors which typically occur during DNA replication. Deficiency in MMR (dMMR) is characterized by mutations and loss of expression in these genes. dMMR is associated with microsatellite instability (MSI), which is defined as a change in the length of a microsatellite in a tumor as compared to normal tissue^{31,32,33}. MSI-high (MSI-H) is a hallmark of Lynch Syndrome (LS), also known as hereditary non-polyposis colorectal cancer, which is caused by germline mutations in MMR genes^{31,34}. LS is associated with an increased risk of developing colorectal cancer, as well as other cancers, including endometrial and stomach cancer^{32,34,35,36}. Specifically, MSH6 mutations are associated with increased risk of ovarian and pancreatic cancer³⁷.

Alterations and prevalence: Somatic mutations in MSH6 are observed in 11% of uterine corpus endometrial carcinoma, 4% colorectal adenocarcinoma, and 3% skin cutaneous melanoma^{8,9}.

Potential relevance: Pembrolizumab (2014) is an anti-PD-1 immune checkpoint inhibitor that is approved for patients with dMMR solid tumors that have progressed on prior therapies³⁸. Nivolumab (2015), an anti-PD-1 immune checkpoint inhibitor, is approved alone

Biomarker Descriptions (continued)

or in combination with the cytotoxic T-lymphocyte antigen 4 (CTLA-4) blocking antibody, ipilimumab (2011), for patients with dMMR colorectal cancer that have progressed on prior treatment^{39,40}.

PALB2 (partner and localizer of BRCA2)

Background: The PALB2 gene encodes the partner and localizer of BRCA2 protein that binds to and promotes intranuclear localization of the breast cancer 2 early onset (BRCA2) protein⁴¹. Also known as FANCN, PALB2 belongs to the Fanconi Anemia (FA) complementation group of proteins that also include FANCA, FANCB, FANCC, FANCD1 (BRCA2), FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL, FANCL, and FANCM. FA genes are tumor suppressors that play a role in interstrand cross-link (ICL) DNA repair through homologous recombination repair (HRR) of double-strand breaks (DSB) and nucleotide excision repair (NER)⁴². Loss of function mutations of genes in the FA family and HRR pathway, including PALB2, can result in the BRCAness phenotype, characterized by a defect in the HRR pathway, mimicking BRCA1 or BRCA2 loss^{6,43}. Germline mutations in FA genes lead to Fanconi Anemia, a condition characterized by chromosomal instability and congenital abnormalities including bone marrow failure and cancer predisposition^{44,45}. Specifically, biallelic germline mutations resulting in PALB2 loss of function confer a predisposition to pediatric malignancies^{46,47}. Additionally, monoallelic germline mutations in PALB2 have been associated with an increased risk of developing breast cancer^{46,48}.

Alterations and prevalence: Somatic alterations in PALB2 include missense or truncating mutations and are observed in 2-6% of melanoma, uterine, bladder, breast, lung, stomach and colorectal cancers⁹.

Potential relevance: The PARP inhibitor, olaparib¹⁰ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes PALB2. In a phase II trial of patients with metastatic, castration-resistant prostate cancer, one patient exhibiting a somatic PALB2 frameshift mutation exhibited durable response to olaparib for 39 weeks^{13,49}. However, olaparib resistance was observed following 9-months of treatment due to the emergence of a secondary deletion which restored the PALB2 reading frame, a resistance mechanism similar to that observed in PARPi treated BRCA mutated patients^{49,50}. In 2022, the FDA granted fast track designation to the small molecule inhibitor, pidnarulex¹⁴, for BRCA1/2, PALB2, or other homologous recombination deficiency (HRD) mutations in breast and ovarian cancers.

SETD2 (SET domain containing 2, histone lysine methyltransferase)

Background: The SETD2 gene encodes the SET domain containing 2 histone lysine methyltransferase, a protein responsible for the trimethylation of lysine-36 on histone H3 (H3K36)^{51,52}. Methylation of H3K36 is a hallmark of active transcription and can be either mono-, di-, or tri-methylated where di- and tri-methylation are thought to be responsible for transcriptional regulation⁵³. Trimethylation of H3K36 by SETD2 promotes post-transcriptional gene silencing and prevents aberrant transcriptional initiation^{54,55}. SETD2 trimethylation activity is also observed to be involved in DNA repair through the recruitment of DNA repair machinery⁵². Specifically, H3K36 tri-methylation by SETD2 has been shown to regulate mismatch repair (MMR) in vivo, wherein the loss of SETD2 results in MMR deficiency (dMMR) and consequent microsatellite instability (MSI)⁵⁶. Both copy number deletion and mutations resulting in SETD2 loss of function have been observed in a variety of cancers, suggesting a tumor suppressor role for SETD2^{52,57}.

Alterations and prevalence: Inactivating somatic mutations in SETD2 were first described in clear cell renal cell carcinoma (ccRCC) and are observed to be predominantly missense or truncating^{9,57,58}. Mutations at codon R1625 are observed to be the most recurrent with R1625C having been identified to result in loss of SETD2 H3K36 trimethylase activity^{9,51}. SETD2 mutation is observed in about 14% of uterine cancer, 12% of ccRCC, 9% of mesothelioma, and 6-7% of melanoma, lung adenocarcinoma, papillary renal cell carcinoma (pRCC), colorectal and bladder cancers⁵¹. Biallelic loss of SETD2 is observed in about 6% of diffuse large B-cell lymphoma, and about 3% of ccRCC and mesothelioma⁵¹.

Potential relevance: Currently, no therapies are approved for SETD2 aberrations. Mutations in SETD2 can be used to support diagnosis of hepatosplenic T-cell lymphoma (HSTCL)⁵⁹.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KRAS Q61E, KRAS L19F

Relevant Therapy	FDA	NCCN	Clinical Trials*
ixazomib, pomalidomide, steroid, cobimetinib	×	×	● (I/II)
AZD-0364	×	×	● (I)
trametinib	×	×	● (I)

ATM D2016Y, ATM E2272K

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	×	×	● (II)
olaparib, ipilimumab + nivolumab, atezolizumab + talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)

PALB2 R753*

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	×	×	● (II)
talazoparib, atezolizumab + talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)

MSH6 C687F

Relevant Therapy	FDA	NCCN	Clinical Trials*
ipilimumab + nivolumab	×	×	● (II)
nivolumab, ipilimumab	×	×	● (I)

MRE11A L485F

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab + talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

KRAS Q61E, KRAS L19F

NCT ID	Title	Phase
NCT03732703	Myeloma-Developing Regimens Using Genomics (MyDRUG) (Genomics Guided Multi-arm Trial of Targeted Agents Alone or in Combination With a Backbone Regimen)	I/II
NCT03091257	An Open-label, Pilot Study of Dabrafenib and/or Trametinib in Patients With Relapsed and/or Refractory Multiple Myeloma	I
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I

ATM D2016Y, ATM E2272K

NCT ID	Title	Phase
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT05071209	A Phase I/II Study of BAY 1895344 (Elimusertib, NSC#810486) in Pediatric Patients With Relapsed or Refractory Solid Tumors	I/II

PALB2 R753*

NCT ID	Title	Phase
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT05071209	A Phase I/II Study of BAY 1895344 (Elimusertib, NSC#810486) in Pediatric Patients With Relapsed or Refractory Solid Tumors	I/II

MSH6 C687F

NCT ID	Title	Phase
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT04500548	3CI Study: Childhood Cancer Combination Immunotherapy. Phase Ib and Expansion Study of Nivolumab Combination Immunotherapy in Children, Adolescent, and Young Adult (CAYA) Patients With Relapsed/Refractory Hypermutant Cancers	I

MRE11A L485F

NCT ID	Title	Phase
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II

Clinical Trials Summary (continued)

MRE11A L485F (continued)

NCT ID	Title	Phase
NCT05071209	A Phase I/II Study of BAY 1895344 (Elimusertib, NSC#810486) in Pediatric Patients With Relapsed or Refractory Solid Tumors	I/II

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSP02, RSP03, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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